

# Assignment - 7

Pappula Kartikeya Raj  
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**Abstract—To understand Inferential Logic in Artificial Intelligence and implement example problems using KANREN, SYMPY and pyDatalog packages in Python.**

## I. DEFINITION OF INFERENTIAL LOGIC IN AI

Inferential Logic in Artificial Intelligence refers to the process of deriving new facts or conclusions from known facts and rules using logical reasoning. It allows AI systems to make decisions, solve problems and generate conclusions based on available knowledge.

In AI, inferential logic is commonly implemented using

- Rule based reasoning
- Predicate logic
- Symbolic reasoning
- Logical programming

Inferential logic mainly uses

- Facts
- Rules
- Inference mechanisms

The system applies inference rules to known facts and produces new conclusions.

## II. PACKAGES USED

### A. Kanren

Used for logic programming and relational reasoning.

### B. Sympy

Used for symbolic mathematics and logical expressions.

### C. pyDatalog

Used for declarative logic programming using facts and rules.

## III. STEPS INVOLVED IN INFERENTIAL LOGIC

### Step 1

Define facts or known information.

### Step 2

Define rules or logical relationships.

### Step 3

Apply inference mechanism.

### Step 4

Derive conclusions from facts and rules.

## IV. EXAMPLES

### A. Example 1

Family Relationship Inference using KANREN

Problem Statement

If John is the father of Mary and Mary is the mother of Sam, infer that John is the grandfather of Sam.

Step 1

Define parent relationships as facts.

Step 2

Define rule for grandparent.

Step 3

Query the system to infer grandfather relationship.

Program:

```
from kanren import Relation, facts, run,
var, conde

# Define relation
parent = Relation()

# Add facts
facts(parent,
      ("John", "Mary"),
      ("Mary", "Sam"))

# Query 1: Father of Mary
x = var()
print("Father of Mary:", run(0, x,
parent(x, "Mary"))))

# Query 2: Grandparent of Sam
x = var()
y = var()

grandparent_goal = (parent(x, y),
parent(y, "Sam"))

print("Grandparent of Sam:", run(0, x,
*grandparent_goal))
```

### Expected Output

Father of Mary: John

Grandparent of Sam: John

### Explanation

The system stores parent facts.

The grandparent rule checks if x is parent of y and y is

parent of z.  
Inference engine derives that John is the grandfather of Sam.

### B. Example 2

Logical Expression Inference using SYMPY

#### Problem Statement

If A implies B and A is true, prove that B is true.

Step 1

Define logical symbols.

Step 2

Define implication expression.

Step 3

Substitute truth values and simplify.

Program

```
from sympy import symbols
from sympy.logic.boolalg import Implies

A, B = symbols('A B')

expr = Implies(A, B)

print("Logical Expression:", expr)

print("Case 1 A=True, B=True:",
      expr.subs({A: True, B: True}))
print("Case 2 A=True, B=False:",
      expr.subs({A: True, B: False}))
```

### C. Example

Student eligibility using pyDatalog

Step 1

Define facts for attendance and result.

Step 2

Define eligibility rule.

Step 3

Query eligibility.

Program

```
from pyDatalog import pyDatalog

pyDatalog.clear()

pyDatalog.create_terms('attendance,
                        passed, eligible, X, A')

# Facts
+ attendance("Ravi", 80)
+ passed("Ravi", True)

+ attendance("Kiran", 60)
```

```
+ passed("Kiran", True)

# Rule
eligible(X) <= (attendance(X, A) & (A >
75) & passed(X, True))

# Query
print("Eligible students:")
print(eligible(X))
```

#### Expected Output

Eligible students:

X = Ravi

#### Explanation

The rule checks attendance greater than 75 and passed status true.  
The inference engine derives that Ravi is eligible but Kiran is not eligible.

## V. RESULT

Inferential logic was successfully studied and implemented using

1. KANREN for relational logic
2. SYMPY for symbolic logical reasoning
3. pyDatalog for rule based inference

The examples demonstrate how AI systems derive conclusions from facts and rules using logical reasoning mechanisms.

## VI. CONCLUSION

Inferential logic plays a major role in Artificial Intelligence. It helps systems reason, make decisions and derive new knowledge from existing information. Using Python packages such as KANREN, SYMPY and pyDatalog, logical inference can be easily implemented in practical applications.

## REFERENCES

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